MAE 155A: Aerospace Engineering Design I

Technical writing

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Overview

The importance of working on your technical writing skills:

- The impact of your work depends heavily on how you communicate it.
- Technical writing is a critical skill for a career in both academia and industry.
- Good technical writing is a habit that takes focused effort to achieve.

The 5 C's of well-written articles:

- Clear: effortless to read because the document flows well and avoids ambiguous sentences.
- Concise: efficient use of words and no wordy sentences.
- Correct: no grammar/spelling errors, prior work cited properly.
- Consistent: adherence to a single set of writing and typesetting conventions throughout.
- Compelling: targeted for the audience and convincing that the reader should care about the message.

Outline

Writing process

Document structuring

Sentence structuring

General recommendations

Writing process

- Write in frequent, short sessions, rather than few, long sessions.
- When writing, focus on the process and not the result. Writing takes time; reward yourself for consistently following the process, not on the quantity or quality of your output (at least early on in the writing process).
- Value getting to a complete first draft even if it needs a lot of work. It may not be great, but it's there. Plan for revision sessions in which your main goal is to cut down to the page limit.
- A suggestion is to write the body of the paper first, followed by the introduction, and then the abstract and conclusion. Make sure to leave extra time to revise the introduction, abstract, and conclusion, as these are the most important (most carefully read) sections of papers.

Recommended steps

1. Active waiting (i.e., planning to write produces ideas)
2. Brainstorming
3. Creating an outline
4. Drafting
5. Revising

Pre-writing—brainstorm

- Write down any and all ideas that come to mind in terms of what you want to include in your article (e.g., key points, findings, tables, figures, equations, data).
- Don’t try to organize your ideas or filter any ideas.
- If you can afford the time, have multiple brainstorming sessions over different days so that you approach each with a fresh perspective.
- I use a spreadsheet to write down my ideas, as that makes it easy to group them into sections later.

Pre-writing—outline

- Now organize your ideas into an outline in bullet form.
- Start with a short and brief, high-level outline and do multiple iterations, getting successively more detailed each time (e.g., sections with a few bullets each).
- Make your final outline three to four levels of bullets deep with individual sentences at the lowest level.
- Going to this much depth in an outline is worthwhile because when working in bullet form, you can stay agile and avoid losing time perfecting wording—rather, you can focus on the actual content and ideas.

Standard Outline for a Technical Paper

Abstract	▪ The problem ▪ How the study addresses this problem ▪ Key results
Introduction	Write this last
Background or Literature Review	The most difficult part of the paper to write
Methods and Materials	Straightforward
Data and Results	Straightforward
Discussion	Your ideas on what the data means
Conclusion	▪ Summary of the findings ▪ Limitations of the study ▪ Recommendations
Acknowledgements	Especially grant sources
References	Pay special attention to the journal guidelines for references

Longo, Julie A. (2014). Technical Writing for Papers and Reports. UNLV.

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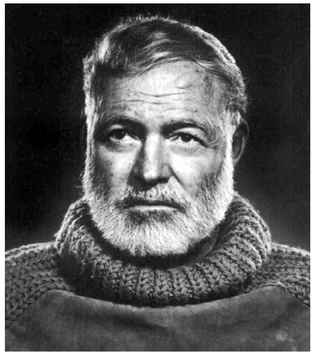
8ste UGent–FirW doctoraatssymposium

Wed 5 Dec 2007 | Het Pand | Gent | Belgium

Plenary session | 17:45–18:30

Modern myths

Shortcomings in scientific writing



Jean-luc Doumont

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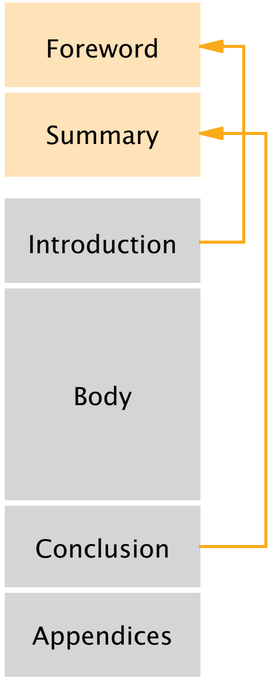
Put what interests readers upfront in an abstract



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An effective summary includes both a *foreword* and a *summary*

Foreword

The *before*

Need

(= problem)

A way to motivate the audience

(what you have ≠ what you want)

Summary

The *after*

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An effective summary includes both a *foreword* and a *summary*

Foreword

The *before*

Need

Task

What I decided or was asked to do in order to address the need

Summary

The *after*

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An effective summary includes both a *foreword* and a *summary*

Foreword

The *before*

Need

Task

Summary

The *after*

Findings

What I found as a result of carrying out the task

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An effective summary includes both a *foreword* and a *summary*

Foreword

The *before*

Need

Task

Summary

The *after*

Findings

Conclusion

(= interpretation/ recommendation)

What the findings mean to you

(possibly what you should do)

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An effective summary includes both a *foreword* and a *summary*

Foreword

The *before*

Need
Task
Object

Summary

The *after*

Findings
Conclusion

What the document does/contains
(and what you should do with it)

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An effective summary includes both a *foreword* and a *summary*

Foreword The *before*	Context Need Task Object	What you need to know to understand the need
Summary The *after*	Findings Conclusion Perspectives	What I/you/they will do in the future

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The summary (*after*) gets messages across

Why Motivation	Context Need Task Object	Why now Why you Why me Why document
What Message(s)	Findings Conclusion Perspectives	What So what What now

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This paper describes PLTO, a link-time instrumentation and optimization tool we have developed for the Intel IA-32 architecture. A number of characteristics of this architecture complicate the task of link-time optimization. These include a large number of op-codes and addressing modes, which increases the complexity of program analysis; variable-length instructions, which complicates disassembly of machine code; a paucity of available registers, which limits the extent of some optimizations; and a reliance on using memory locations for holding values and for parameter passing, which complicates program analysis and optimization. We describe how PLTO addresses these problems and the resulting performance improvements it is able to achieve.

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Context
Need
Task
Object

Findings
Conclusion
Perspectives

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On simple RISC architectures, post-link-time optimization of executable programs delivers significant performance improvements. However, the applicability of this technique has not yet been evaluated for more complex CISC architectures such as the widely used Intel IA-32 processor family. We have developed PLTO, a link-time instrumentation and optimization tool for IA-32. This paper describes how PLTO addresses the complexities of this processor architecture and which analyses and optimizations contribute to the achieved performance improvements. Currently, PLTO achieves a moderate speedup of about 6% on average. We expect bigger speedups once we have solved a remaining problem involving significantly increased instruction cache misses.

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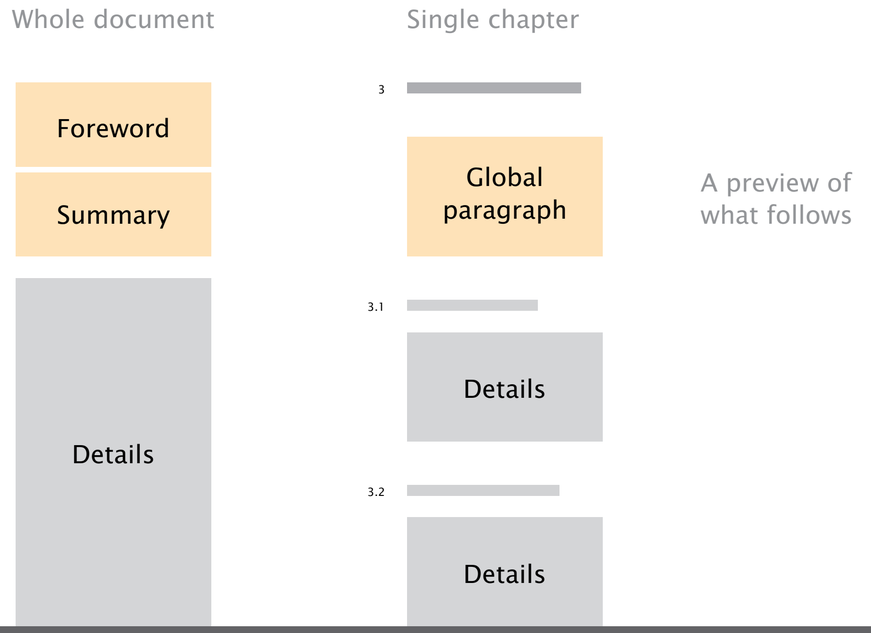
Apply the theorem-proof approach recursively



A preview of what follows

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Single-use, disposable medical devices are pre-packaged and sterilized by the manufacturer. The packaging is chosen to provide protection for the product, to facilitate sterilization, to maintain sterility, and to be easy to use. Reusable devices, by contrast, must be durable both in service and in their ability to withstand repeated sterilization. …

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Single-use, disposable medical devices are pre-packaged and sterilized by the manufacturer. The packaging is chosen to provide protection for the product, to facilitate sterilization, to maintain sterility, and to be easy to use. Reusable devices, **by contrast,** must be durable both in service and in their ability to withstand repeated sterilization. …

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Medical devices may be broadly divided into two categories, disposable and reusable, having different sterilization requirements. Single-use, disposable medical devices are pre-packaged and sterilized by the manufacturer. The packaging is chosen to provide protection for the product, to facilitate sterilization, to maintain sterility, and to be easy to use. Reusable devices, by contrast, must be durable both in service and in their ability to withstand repeated sterilization. …

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Figure 2 shows the evolution of the germanium content in the SiGe layer. Obviously there is a nearly linear decrease of the germanium content with increasing fluence, apart from a near constant region around 2×10^{17} at/cm^2. Knowing the number of atoms that has been sputtered and the O_2 fluence, it is then quite easy to calculate the germanium sputter yield.

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To be kept Ineffectively redundant

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The germanium content decreases linearly with increasing fluence apart from a near constant region around 2×10^{17} at/cm^2 (Figure 2). Knowing the number of atoms that has been sputtered and the O_2 fluence, it is then quite easy to calculate the germanium sputter yield.

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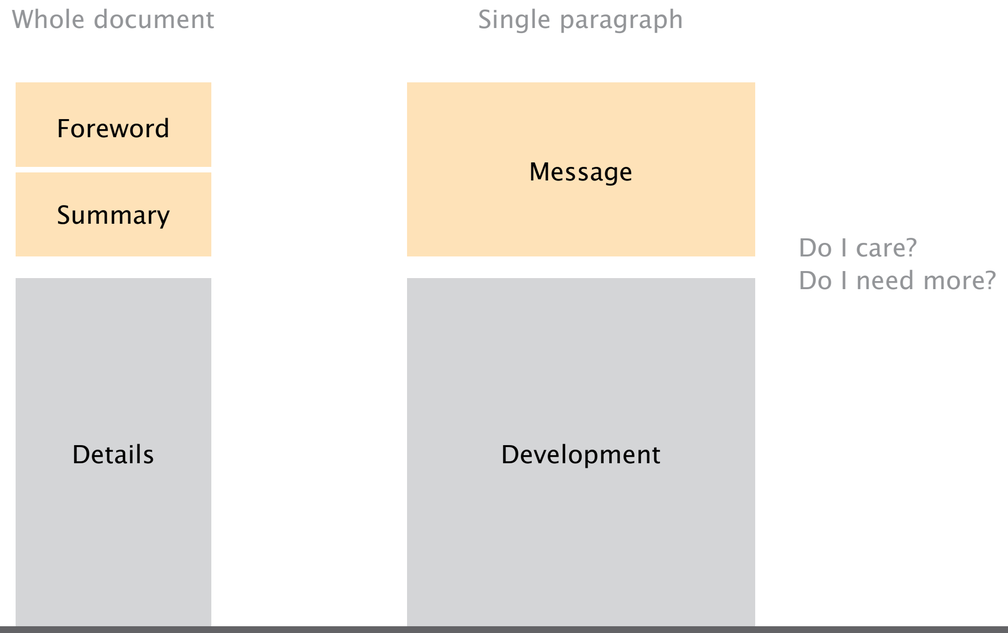
Again, state your message upfront



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Document structuring

- The goal is to make the document read effortlessly, so as to maximize the signal-to-noise ratio. The signal is the information and messages we want to communicate, and noise could include ambiguity, unclear section organization, or abrupt transitions.
- It helps to end the introduction with a roadmap for the document, e.g., “This document proceeds as follows. Section 2 describes the …”
- It also helps to frequently remind the reader what was discussed previously and what is to come. Start sections with a brief description of the context. For instance, “With the background now established, this section describes…”
- Don’t give previews (e.g., “The results will be discussed later”); rather, use a punchline (e.g., “The method improves the result by X %”).

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In Figure 3, it can be seen that a very good agreement is achieved between …

"it can be seen that" site:ugent.be 600 hits

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Main clause

In Figure 3, it can be seen that a very good agreement is achieved between …

Main information

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When writing a complex sentence, place the main information in the main clause

First, suppress the main clause

~~In Figure 3, it can be seen that~~ a very good agreement is achieved between …

Then, put back what you may have lost

A very good agreement is achieved between experimental and simulation results (Figure 3).

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Express the action with a verb, not with a noun

Experimental and simulation results agree very well (Figure 3).

A very good agreement is achieved between experimental and simulation results (Figure 3).

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Suppress unnecessary words

Experimental and simulation results agree ~~very~~ well (Figure 3).

?

Experimental and simulation results agree (Figure 3).

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Before

In Figure 3, it can be seen that a very good agreement is achieved between experimental and simulation results.

After

Experimental and simulation results agree (Figure 3).

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This region is believed to be the source of many short-period comets.

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Who believes this?

The scientific community?

The authors?

This region is believed to be the source of many short-period comets.

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Identify people (when people matter) in whichever way is accurate

In the third person

Astronomers believe this region to be the source …
Asimov *et al.* believe this region to be … [7]

In the first person

We believe this region to be the source of …
Our observations suggest this region to be …

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For maximum impact, use *we* to mean *we, the authors* only

Avoid *we* for communication purposes
Focus on the document (*Section 3.2 presents*)

Avoid *we* to designate everyone
Use *one*—or change the construction

Avoid *we* to designate a single author
or say who else is included (*My supervisor and I*)

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Sentence structuring

- Remember that the hierarchy is, in order: sentences, clauses, phrases, and words.
- Present old or familiar information early in sentences, and present new and unfamiliar information later in sentences.
- Minimize sentences with multiple clauses. Break up long sentences into smaller sentences. Paragraphs with compact, clear, and direct sentences are much more impactful and convincing than paragraphs with long sentences that take a while to get to the point.
- A good rule of thumb is that if you need to take a breath in the middle of the sentence when reading it aloud, it is too long.
- Place key information in the main clause.
- Original: It can be seen that experimental and simulation results agree.
- Improved: Experimental and simulation results agree.

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General recommendations

General recommendations (1/3)

- Use a direct and factual tone, rather than an opinionated one. Use active voice—frequent use of first-person plural (‘we’) is preferred over passive voice, but even better is using active voice without using ‘we’.
- Use past tense for narrative descriptions of experiments that were performed; otherwise, use present tense.
- Avoid relative qualifiers that are non-specific, such as very, little, really, etc. Try to use specific, quantitative statements.
- Clarify mathematical definitions using examples.
- Avoid contractions (e.g., ‘can’t’).
- Use vector graphics for figures (pdf) or high-resolution images.
- Figure labels should go below, and table labels should go above.

General recommendations (2/3)

- Spell out an acronym only the first time it is used (e.g., “Multidisciplinary design optimization (MDO)”).
- Compound modifiers must be hyphenated (e.g., “a large-scale optimization problem”). The same goes for numbers and units (e.g., “a 5-in-radius propeller”).
- When using units without quantities, spell them out (e.g., “Figure 1 shows range in meters”).
- The em dash (three dashes in LaTeX) is a punctuation mark as wide as an “M” when typeset, and it can replace commas, parentheses, colons, and semicolons. When used in place of parentheses, there should be no space on either side (e.g., “The algorithm—which is provably convergent—yields …”). The en dash (two dashes in LaTeX) is used to indicate a range (e.g., “2020–2021”) or to hyphenate proper nouns (e.g., “the Newton–Raphson method”).

General recommendations (3/3)

- In Latin, ‘i.e.’ means “that is” and ‘e.g.’ means “for example.” They should be followed by a comma.
- Use ‘that’ for restrictive (essential) phrases and ‘which’ for nonrestrictive (nonessential) phrases. ‘Which’ is commonly used incorrectly. As a rule of thumb, use ‘that’ unless ‘which’ is clearly more appropriate.
- Use ‘different’ to refer to two dissimilar objects and ‘various’ to refer to several types of items.
- With prefixes, include a hyphen if there is a doubling of letters (e.g., “co-orbit”) or if the compound term is new; otherwise, drop the hyphen (e.g., “coexist”).
- Use a comma when joining independent clauses with one of the following conjunctions: and, or, nor, but.
- Do not start a sentence with a number.

Mathematics

- Use words to spell out numbers 1-9 and numerals for 10 and above (e.g., “eight propellers and 16 blades”).
- Use numerals for dates, numbers in a series, and ages. When expressing a number with units, include a single space (e.g., “15 m”).
- Equations should be treated as part of sentences. The preceding text should lead into the equation; therefore, the text immediately preceding an equation should not end with a period. Equations should be followed by a comma or period, as follows:

  $$F = ma.$$

  Any text that follows should not be indented, unless it begins a new idea.
- With large numbers, use a space to separate from the final three digits (e.g., “60 000”).
- Refer to equations by number only (e.g., “in (1)”), except when leading off a sentence (e.g., “Equation (1) defines …”).

Redundant words

(already) existing	never (before)
at (the) present (time)	none (at all)
(basic) fundamentals	now (at this time)
(completely) eliminate	period (of time)
(continue to) remain	(private) industry
currently (being)	(separate) entities
(currently) underway	start (out)
(empty) space	write (out)
had done (previously)	(still) persists
introduced (a new)	
mix (together)	

Kelley, N. (2023). Sentence structure of technical writing. Program in Writing and Humanistic Studies.